OM Industrial Products Corp. — Carbon Steel Hydraulic Fittings

Specification pack, 10 series · generated 16 Sep 2026 from OMi_Fitting_Library_MasterData_v1 / 04_DESIGN_TABLES

Series	Description	Parts	Verified	Assumed	Missing	Master model
2404	Male Connector (JIC 37° male × NPTF male)	54	26	28	0	MASTER BUILT (hand)
6400	Straight Thread Connector (JIC 37° male × SAE ORB male)	65	9	49	7	MASTER BUILT (hand)
2403	Male Union (JIC 37° male × JIC 37° male)	33	13	20	0	MASTER BUILT (hand + macro)
2405	Female Connector (JIC 37° male × NPTF female)	32	16	16	0	MASTER BUILT (hand + macro)
2500	90° Union Elbow (JIC 37° male × JIC 37° male)	27	17	10	0	TABLED - macro drafted
2502	90° Female Pipe Elbow (JIC 37° male × NPTF female)	29	9	20	0	TABLED
2503	45° Male Elbow (JIC 37° male × NPTF male)	60	18	42	0	TABLED
2603	Union Tee (JIC 37° male × 3)	41	19	22	0	TABLED
5406	Pipe Reducer Bushing (NPTF male × NPTF female)	31	0	31	0	TABLED
6803	Branch Tee (JIC 37° male × JIC 37° male × SAE ORB branch)	34	13	21	0	TABLED
TOTAL		**406**	**140**	**259**	**7**	

How to read the status column. VERIFIED = two independent sources agree (Air-Way catalog plus a military/SAE standard sheet or a Brennan drawing/STEP measurement). ASSUMED = one source only (Air-Way catalog) — accurate to the catalog, not yet independently confirmed; per the project rule these are not released for production drawings. MISSING = no source, or sources conflict; the part stays in scope but is not modeled until resolved. No value in this pack was estimated or interpolated.

Standards applied. JIC 37° male ends: SAE J514 (MS33656J geometry, verified against Brennan STEP measurements). NPTF: ASME B1.20.3 (tap drill per Air-Way). SAE O-ring boss: SAE J1926 / AS4395 (groove per Parker ORD-5700 Table 4-4). Pipe-thread hex and lengths: MS51503 / MS51506 / MS51508 / MS51510 / MS51521 where a matched size exists. Material: carbon steel. Dimensions in inches. Thread callouts are nominal (UN/UNF-2A, NPTF).

2404 Male Connector (JIC 37° male × NPTF male)

54 part numbers · MASTER BUILT (hand) · source table: DesignTable_2404_FINAL.xlsx

Part number	Thread(s)	L overall	Hex A/F	Bore	Pipe OD	NPTF end L	Status	Source / note
2404-02-02	1/8-27 × 5/16-24	1.11	0.4375	0.062	0.405	0.38	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-03-02	1/8-27 × 3/8-24	1.14	0.4375	0.125	0.405	0.38	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-04-02	1/8-27 × 7/16-20	1.22	0.5	0.172	0.405	0.38	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-04-04	1/4-18 × 7/16-20	1.42	0.5625	0.172	0.54	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-04-06	3/8-18 × 7/16-20	1.42	0.75	0.17	0.675	0.56	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-04-08	1/2-14 × 7/16-20	1.68	0.88	0.17	0.84	0.75	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-04-12	3/4-14 × 7/16-20	1.75	1.13	0.17	1.05	0.75	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-05-02	1/8-27 × 1/2-20	1.22	0.5625	0.188	0.405	0.38	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-05-04	1/4-18 × 1/2-20	1.42	0.5625	0.234	0.54	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-05-06	3/8-18 × 1/2-20	1.42	0.75	0.23	0.675	0.56	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-02	1/8-27 × 9/16-18	1.24	0.625	0.188	0.405	0.38	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-04	1/4-18 × 9/16-18	1.43	0.625	0.281	0.54	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-06	3/8-18 × 9/16-18	1.43	0.75	0.297	0.675	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-08	1/2-14 × 9/16-18	1.69	0.875	0.297	0.84	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-12	3/4-14 × 9/16-18	1.75	1.13	0.3	1.05	0.75	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-06-16	1-11 1/2 × 9/16-18	1.94	1.38	0.3	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-08-04	1/4-18 × 3/4-16	1.53	0.8125	0.281	0.54	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-08-06	3/8-18 × 3/4-16	1.53	0.8125	0.391	0.675	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-08-08	1/2-14 × 3/4-16	1.79	0.875	0.391	0.84	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-08-12	3/4-14 × 3/4-16	1.85	1.125	0.391	1.05	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-08-16	1-11 1/2 × 3/4-16	2.05	1.38	0.39	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-10-04	1/4-18 × 7/8-14	1.7	0.94	0.28	0.54	0.56	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-10-06	3/8-18 × 7/8-14	1.7	0.9375	0.406	0.675	0.56	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-10-08	1/2-14 × 7/8-14	1.89	0.9375	0.484	0.84	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-10-12	3/4-14 × 7/8-14	1.95	1.13	0.48	1.05	0.75	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-10-16	1-11 1/2 × 7/8-14	2.15	1.38	0.48	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-12-06	3/8-18 × 1 1/16-12	1.87	1.13	0.41	0.675	0.56	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-12-08	1/2-14 × 1 1/16-12	2.06	1.125	0.531	0.84	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-12-12	3/4-14 × 1 1/16-12	2.06	1.125	0.609	1.05	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-12-16	1-11 1/2 × 1 1/16-12	2.25	1.38	0.61	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)

Part number	Thread(s)	L overall	Hex A/F	Bore	Pipe OD	NPTF end L	Status	Source / note
2404-12-20	1 1/4-11 1/2 × 1 1/16-12	2.36	1.69	0.61	1.66	0.97	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-12-24	1 1/2-11 1/2 × 1 1/16-12	2.46	2	0.61	1.9	1	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-16-12	3/4-14 × 1 5/16-12	2.11	1.375	0.719	1.05	0.75	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-16-16	1-11 1/2 × 1 5/16-12	2.3	1.375	0.844	1.315	0.94	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-16-20	1 1/4-11 1/2 × 1 5/16-12	2.4	1.69	0.84	1.66	0.97	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-16-24	1 1/2-11 1/2 × 1 5/16-12	2.51	2	0.84	1.9	1	VERIFIED	RESOLVED: L=2.51 - Air-Way+Brennan drawing agree on 2.51 (Parker 2.50 outvoted 2-1, 2026-09-05)
2404-20-12	3/4-14 × 1 5/8-12	2.23	1.69	0.72	1.05	0.75	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-20-16	1-11 1/2 × 1 5/8-12	2.42	1.69	0.94	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-20-20	1 1/4-11 1/2 × 1 5/8-12	2.45	1.6875	1.078	1.66	0.97	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-20-24	1 1/2-11 1/2 × 1 5/8-12	2.55	2	1.08	1.9	1	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-20-32	2-11 1/2 × 1 5/8-12	2.73	2.5	1.08	2.375	1.03	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-24-16	1-11 1/2 × 1 7/8-12	2.62	2	0.94	1.315	0.94	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-24-20	1 1/4-11 1/2 × 1 7/8-12	2.65	2	1.25	1.66	0.97	VERIFIED	RESOLVED: L=2.65 - Air-Way+Brennan drawing agree on 2.65 (Parker 2.66 outvoted 2-1, 2026-09-05)
2404-24-24	1 1/2-11 1/2 × 1 7/8-12	2.68	2	1.312	1.9	1	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-24-32	2-11 1/2 × 1 7/8-12	2.86	2.63	1.31	2.375	1.03	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-32-24	1 1/2-11 1/2 × 2 1/2-12	3.08	2.63	1.5	1.9	1	ASSUMED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-32-32	2-11 1/2 × 2 1/2-12	3.11	2.625	1.781	2.375	1.03	VERIFIED	Brennan STEP measured == MS33656J published (VERIFIED)
2404-02-04	1/4-18 NPTF × 5/16-24	1.33	0.562	0.062	0.54	0.56	ASSUMED	BLOCKED: CB depth unsourced
2404-08-02	1/8-27 NPTF × 3/4-16	1.34	0.812	0.188	0.405	0.38	ASSUMED	BLOCKED: CB depth unsourced
2404-16-32	2-11 1/2 NPTF × 1 5/16-12	2.69	2.5	0.844	2.375	1.03	ASSUMED	BLOCKED: CB depth unsourced
2404-32-20	1 1/4-11 1/2 NPTF × 2 1/2-12	3.05	2.625	1.25	1.66	0.97	ASSUMED	BLOCKED: CB depth unsourced
2404-32-48	3-8 NPTF × 2 1/2-12	3.74	3.75	1.781	3.5	—	ASSUMED	BLOCKED: NPTF -48 end length unmeasured; CB depth unsourced
2404-40-32	2-11 1/2 NPTF × 3-12	2.89	3	1.938	2.375	1.03	VERIFIED	Brennan product drawing 2026-09-05 (OMi-only SKU; absent from Air-Way) ; L matches Brennan STEP (VERIFIED); CB depth 1.5975 measured
2404-40-40	2 1/2-8 NPTF × 3-12	3.37	3.25	2.318	2.875	—	ASSUMED	BLOCKED: NPTF -40 end length unmeasured

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

6400 Straight Thread Connector (JIC 37° male × SAE ORB male)

65 part numbers · MASTER BUILT (hand) · source table: DesignTable_6400.xlsx

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	ORB thread Ø	ORB groove Ø	ORB groove W	Status	Source / note
6400-02-02-O	5/16-24 × 5/16-24	1.06	0.4375	0.06	0.06	0.234	0.25	0.063	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-03-03-O	3/8-24 × 3/8-24	1.1	0.5	0.13	0.13	0.234	0.312	0.063	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-02-O	5/16-24 × 7/16-20	1.17	0.5	0.06	0.17	0.234	0.25	0.063	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-03-O	3/8-24 × 7/16-20	1.17	0.5	0.13	0.17	0.234	0.312	0.063	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-04-O	7/16-20 × 7/16-20	1.23	0.5625	0.17	0.17	0.265	0.364	0.075	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-05-O	1/2-20 × 7/16-20	1.23	0.625	0.23	0.17	0.265	0.426	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-06-O	9/16-18 × 7/16-20	1.29	0.6875	0.3	0.17	0.312	0.481	0.083	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-08-O	3/4-16 × 7/16-20	1.37	0.875	0.39	0.17	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-10-O	7/8-14 × 7/16-20	1.49	1	0.48	0.17	0.391	0.773	0.107	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-04-12-O	1 1/16-12 × 7/16-20	1.66	1.25	0.61	0.17	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-05-04-O	7/16-20 × 1/2-20	1.23	0.5625	0.17	0.23	0.265	0.364	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-05-05-O	1/2-20 × 1/2-20	1.23	0.625	0.23	0.23	0.265	0.426	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-05-06-O	9/16-18 × 1/2-20	1.29	0.6875	0.3	0.23	0.312	0.481	0.083	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-05-08-O	3/4-16 × 1/2-20	1.37	0.875	0.39	0.23	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-04-O	7/16-20 × 9/16-18	1.27	0.625	0.17	0.3	0.265	0.364	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-05-O	1/2-20 × 9/16-18	1.27	0.625	0.23	0.3	0.265	0.426	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-06-O	9/16-18 × 9/16-18	1.3	0.6875	0.3	0.3	0.312	0.481	0.083	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-08-O	3/4-16 × 9/16-18	1.38	0.875	0.3	0.3	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-10-O	7/8-14 × 9/16-18	1.5	1	0.48	0.3	0.391	0.773	0.107	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	ORB thread Ø	ORB groove Ø	ORB groove W	Status	Source / note
6400-06-12-O	1 1/16-12 × 9/16-18	1.66	1.25	0.61	0.3	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-06-16-O	1 5/16-12 × 9/16-18	1.68	1.5	0.84	0.3	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-08-04-O	7/16-20 × 3/4-16	1.49	0.8125	0.17	0.39	0.265	0.364	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-05-O	1/2-20 × 3/4-16	1.49	0.8125	0.23	0.39	0.265	0.426	0.075	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-06-O	9/16-18 × 3/4-16	1.44	0.8125	0.3	0.39	0.312	0.481	0.083	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-08-O	3/4-16 × 3/4-16	1.48	0.875	0.39	0.39	0.344	0.66	0.094	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-10-O	7/8-14 × 3/4-16	1.6	1	0.48	0.39	0.391	0.773	0.107	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-12-O	1 1/16-12 × 3/4-16	1.76	1.25	0.61	0.39	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-08-14-O	1 3/16-12 × 3/4-16	1.76	1.375	0.72	0.39	0.469	1.07	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-08-16-O	1 5/16-12 × 3/4-16	1.79	1.5	0.84	0.39	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-08-20-O	1 5/8-12 × 3/4-16	1.87	1.875	1.08	0.39	0.469	1.507	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-10-06-O	9/16-18 × 7/8-14	1.71	0.9375	0.3	0.48	0.312	0.481	0.083	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-10-08-O	3/4-16 × 7/8-14	1.64	0.9375	0.39	0.48	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-10-10-O	7/8-14 × 7/8-14	1.7	1	0.48	0.48	0.391	0.773	0.107	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-10-12-O	1 1/16-12 × 7/8-14	1.86	1.25	0.48	0.48	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-10-16-O	1 5/16-12 × 7/8-14	1.89	1.5	0.84	0.48	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-12-06-O	9/16-18 × 1 1/16-12	1.86	1.125	0.3	0.61	0.312	0.481	0.083	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-12-08-O	3/4-16 × 1 1/16-12	1.94	1.125	0.39	0.61	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-12-10-O	7/8-14 × 1 1/16-12	1.88	1.125	0.48	0.61	0.391	0.773	0.107	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	ORB thread Ø	ORB groove Ø	ORB groove W	Status	Source / note
6400-12-12-O	1 1/16-12 × 1 1/16-12	1.97	1.25	0.61	0.61	0.469	0.945	0.125	VERIFIED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-12-14-O	1 3/16-12 × 1 1/16-12	1.96	1.375	0.72	0.61	0.469	1.07	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-12-16-O	1 5/16-12 × 1 1/16-12	1.99	1.5	0.84	0.61	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-12-20-O	1 5/8-12 × 1 1/16-12	2.08	1.875	1.08	0.61	0.469	1.507	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-12-24-O	1 7/8-12 × 1 1/16-12	2.15	2.125	1.31	0.61	0.469	1.756	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-14-12-O	1 1/16-12 × 1 3/16-12	1.99	1.25	0.61	0.72	0.469	0.945	0.125	ASSUMED	JIC: Brennan fleet measured (no MS33656 row; ASSUMED); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-14-14-O	1 3/16-12 × 1 3/16-12	1.99	1.375	0.72	0.72	0.469	1.07	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-14-16-O	1 5/16-12 × 1 3/16-12	2.02	1.5	0.84	0.72	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-16-08-O	3/4-16 × 1 5/16-12	2.01	1.375	0.39	0.84	0.344	0.66	0.094	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-16-10-O	7/8-14 × 1 5/16-12	2.07	1.375	0.48	0.84	0.391	0.773	0.107	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-16-12-O	1 1/16-12 × 1 5/16-12	2.04	1.375	0.61	0.84	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)
6400-16-14-O	1 3/16-12 × 1 5/16-12	2.04	1.375	0.72	0.84	0.469	1.07	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-16-16-O	1 5/16-12 × 1 5/16-12	2.04	1.5	0.84	0.84	0.469	1.195	0.125	VERIFIED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-16-20-O	1 5/8-12 × 1 5/16-12	2.12	1.875	1.08	0.84	0.469	1.507	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-16-24-O	1 7/8-12 × 1 5/16-12	2.2	2.125	1.31	0.84	0.469	1.756	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-20-12-O	1 1/16-12 × 1 5/8-12	2.3	1.6875	0.61	1.08	0.469	0.945	0.125	ASSUMED	JIC: VERIFIED (MS33656J = Brennan fleet); ORB thd: MS51527 J2 transfer (ASSUMED)

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	ORB thread Ø	ORB groove Ø	ORB groove W	Status	Source / note
6400-20-16-O	1 5/16-12 × 1 5/8-12	2.33	1.6875	0.84	1.08	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-20-20-O	1 5/8-12 × 1 5/8-12	2.17	1.875	1.08	1.08	0.469	1.507	0.125	VERIFIED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-20-24-O	1 7/8-12 × 1 5/8-12	2.24	2.125	1.31	1.08	0.469	1.756	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-20-32-O	2 1/2-12 × 1 5/8-12	2.4	2.75	1.78	1.08	0.469	2.381	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-24-16-O	1 5/16-12 × 1 7/8-12	2.53	2	0.84	1.31	0.469	1.195	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-24-20-O	1 5/8-12 × 1 7/8-12	2.53	2	1.08	1.31	0.469	1.507	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-24-24-O	1 7/8-12 × 1 7/8-12	2.37	2.125	1.31	1.31	0.469	1.756	0.125	VERIFIED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-24-32-O	2 1/2-12 × 1 7/8-12	2.53	2.75	1.78	1.31	0.469	2.381	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config
6400-32-20-O	1 5/8-12 × 2 1/2-12	2.94	2.625	1.08	1.78	0.469	1.507	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400-32-20-O STEP (ASSUMED, single source; method validated on -4/-12)
6400-32-24-O	1 7/8-12 × 2 1/2-12	2.94	2.625	1.31	1.78	0.469	1.756	0.125	ASSUMED	ORB_GRVW measured from Brennan 6400 STEP (ASSUMED, single source; method validated on -4/-12 pub values)
6400-32-32-O	2 1/2-12 × 2 1/2-12	2.78	2.75	1.78	1.78	0.469	2.381	—	MISSING	BLOCKED - ORB groove width unpublished for this boss dash; row intentionally EXCLUDED from DesignTable_6400_SW2.xlsx so no SolidWorks config

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2403 Male Union (JIC 37° male × JIC 37° male)

33 part numbers · MASTER BUILT (hand + macro) · source table: DesignTable_2403.xlsx

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	Status	Source / note
2403-02-02	5/16-24 × 5/16-24	1.17	0.44	0.06	0.06	VERIFIED	MS51501 L1/hex/bore AGREE
2403-03-03	3/8-24 × 3/8-24	1.23	0.44	0.13	0.13	VERIFIED	MS51501 L1/hex/bore AGREE
2403-04-03	7/16-20 × 3/8-24	1.3	0.5	0.17	0.13	ASSUMED	unequal bores - bore step location unsourced
2403-04-04	7/16-20 × 7/16-20	1.37	0.5	0.17	0.17	VERIFIED	MS51501 L1/hex/bore AGREE
2403-05-04	1/2-20 × 7/16-20	1.37	0.56	0.23	0.17	ASSUMED	unequal bores - bore step location unsourced
2403-05-05	1/2-20 × 1/2-20	1.37	0.56	0.23	0.23	VERIFIED	MS51501 L1/hex/bore AGREE
2403-06-04	9/16-18 × 7/16-20	1.4	0.63	0.3	0.17	ASSUMED	unequal bores - bore step location unsourced
2403-06-05	9/16-18 × 1/2-20	1.4	0.63	0.3	0.23	ASSUMED	unequal bores - bore step location unsourced
2403-06-06	9/16-18 × 9/16-18	1.41	0.63	0.3	0.3	VERIFIED	MS51501 L1/hex/bore AGREE
2403-08-04	3/4-16 × 7/16-20	1.51	0.81	0.39	0.17	ASSUMED	unequal bores - bore step location unsourced
2403-08-06	3/4-16 × 9/16-18	1.52	0.81	0.39	0.3	ASSUMED	unequal bores - bore step location unsourced
2403-08-08	3/4-16 × 3/4-16	1.62	0.81	0.39	0.39	VERIFIED	MS51501 L1/hex/bore AGREE
2403-10-06	7/8-14 × 9/16-18	1.68	0.94	0.48	0.3	ASSUMED	unequal bores - bore step location unsourced
2403-10-08	7/8-14 × 3/4-16	1.78	0.94	0.48	0.39	ASSUMED	unequal bores - bore step location unsourced
2403-10-10	7/8-14 × 7/8-14	1.88	0.94	0.48	0.48	VERIFIED	MS51501 L1/hex/bore AGREE
2403-12-04	1 1/16-12 × 7/16-20	1.85	1.13	0.61	0.17	ASSUMED	unequal bores - bore step location unsourced
2403-12-06	1 1/16-12 × 9/16-18	1.85	1.13	0.61	0.3	ASSUMED	unequal bores - bore step location unsourced
2403-12-08	1 1/16-12 × 3/4-16	1.95	1.13	0.61	0.39	ASSUMED	unequal bores - bore step location unsourced
2403-12-10	1 1/16-12 × 7/8-14	2.05	1.13	0.61	0.48	ASSUMED	unequal bores - bore step location unsourced
2403-12-12	1 1/16-12 × 1 1/16-12	2.16	1.13	0.61	0.61	VERIFIED	MS51501 L1/hex/bore AGREE
2403-14-12	1 3/16-12 × 1 1/16-12	2.18	1.25	0.72	0.61	ASSUMED	unequal bores - bore step location unsourced
2403-14-14	1 3/16-12 × 1 3/16-12	2.21	1.25	0.72	0.72	VERIFIED	MS51501 L1/hex/bore AGREE
2403-16-08	1 5/16-12 × 3/4-16	2	1.38	0.84	0.39	ASSUMED	unequal bores - bore step location unsourced
2403-16-12	1 5/16-12 × 1 1/16-12	2.2	1.38	0.84	0.61	ASSUMED	unequal bores - bore step location unsourced
2403-16-16	1 5/16-12 × 1 5/16-12	2.25	1.38	0.84	0.84	VERIFIED	MS51501 L1/hex/bore AGREE
2403-20-12	1 5/8-12 × 1 1/16-12	2.34	1.69	1.08	0.61	ASSUMED	unequal bores - bore step location unsourced
2403-20-16	1 5/8-12 × 1 5/16-12	2.38	1.69	1.08	0.84	ASSUMED	unequal bores - bore step location unsourced
2403-20-20	1 5/8-12 × 1 5/8-12	2.43	1.69	1.08	1.08	VERIFIED	MS51501 L1/hex/bore AGREE
2403-24-12	1 7/8-12 × 1 1/16-12	2.53	2	1.31	0.61	ASSUMED	unequal bores - bore step location unsourced
2403-24-20	1 7/8-12 × 1 5/8-12	2.62	2	1.31	1.08	ASSUMED	unequal bores - bore step location unsourced

Part number	Thread(s)	L overall	Hex A/F	Bore A	Bore B	Status	Source / note
2403-24-24	1 7/8-12 × 1 7/8-12	2.75	2	1.31	1.31	VERIFIED	MS51501 L1/hex/bore AGREE
2403-32-24	2 1/2-12 × 1 7/8-12	3.15	2.63	1.78	1.31	ASSUMED	unequal bores - bore step location unsourced
2403-32-32	2 1/2-12 × 2 1/2-12	3.4	2.63	1.78	1.78	VERIFIED	MS51501 L1/hex/bore AGREE

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2405 Female Connector (JIC 37° male × NPTF female)

32 part numbers · MASTER BUILT (hand + macro) · source table: DesignTable_2405.xlsx

Part number	Thread(s)	L overall	Hex A/F	Tap drill Ø	Pipe thread depth	Bore	Status	Source / note
2405-02-02	1/8-27 NPTF (female) × 5/16-24	1.12	0.56	0.33	0.46	0.06	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-03-04	1/4-18 NPTF (female) × 3/8-24	1.32	0.75	0.43	0.67	0.13	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-04-02	1/8-27 NPTF (female) × 7/16-20	1.19	0.56	0.33	0.46	0.17	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-04-04	1/4-18 NPTF (female) × 7/16-20	1.39	0.75	0.43	0.67	0.17	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-04-06	3/8-18 NPTF (female) × 7/16-20	1.45	0.88	0.57	0.68	0.17	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-04-08	1/2-14 NPTF (female) × 7/16-20	1.68	1.13	0.7	0.9	0.17	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-05-02	1/8-27 NPTF (female) × 1/2-20	1.17	0.56	0.33	0.46	0.23	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-05-04	1/4-18 NPTF (female) × 1/2-20	1.39	0.75	0.43	0.67	0.23	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-05-06	3/8-18 NPTF (female) × 1/2-20	1.45	0.88	0.57	0.68	0.23	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-06-02	1/8-27 NPTF (female) × 9/16-18	1.19	0.63	0.33	0.46	0.3	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-06-04	1/4-18 NPTF (female) × 9/16-18	1.4	0.75	0.43	0.67	0.3	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-06-06	3/8-18 NPTF (female) × 9/16-18	1.46	0.88	0.57	0.68	0.3	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-06-08	1/2-14 NPTF (female) × 9/16-18	1.69	1.13	0.7	0.9	0.3	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-08-04	1/4-18 NPTF (female) × 3/4-16	1.55	0.81	0.43	0.67	0.39	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-08-06	3/8-18 NPTF (female) × 3/4-16	1.56	0.88	0.57	0.68	0.39	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-08-08	1/2-14 NPTF (female) × 3/4-16	1.79	1.13	0.7	0.9	0.39	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-08-12	3/4-14 NPTF (female) × 3/4-16	1.85	1.38	0.91	0.91	0.39	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-10-08	1/2-14 NPTF (female) × 7/8-14	1.89	1.13	0.7	0.9	0.48	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-10-12	3/4-14 NPTF (female) × 7/8-14	1.95	1.38	0.91	0.91	0.48	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-12-08	1/2-14 NPTF (female) × 1 1/16-12	2.05	1.13	0.7	0.9	0.61	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-12-12	3/4-14 NPTF (female) × 1 1/16-12	2.06	1.38	0.91	0.91	0.61	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-12-16	1-11 1/2 NPTF (female) × 1 1/16-12	2.3	1.63	1.14	1.14	0.61	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-16-12	3/4-14 NPTF (female) × 1 5/16-12	2.12	1.38	0.91	0.91	0.84	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way

Part number	Thread(s)	L overall	Hex A/F	Tap drill Ø	Pipe thread depth	Bore	Status	Source / note
2405-16-16	1-11 1/2 NPTF (female) × 1 5/16-12	2.35	1.63	1.14	1.14	0.84	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-16-20	1 1/4-11 1/2 NPTF (female) × 1 5/16-12	2.44	2	1.48	1.16	0.84	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-20-16	1-11 1/2 NPTF (female) × 1 5/8-12	2.47	1.69	1.14	1.14	1.08	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-20-20	1 1/4-11 1/2 NPTF (female) × 1 5/8-12	2.49	2	1.48	1.16	1.08	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-20-24	1 1/2-11 1/2 NPTF (female) × 1 5/8-12	2.49	2.38	1.72	1.16	1.08	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-24-24	1 1/2-11 1/2 NPTF (female) × 1 7/8-12	2.62	2.38	1.72	1.16	1.31	VERIFIED	MS51503D Amd1 L2/hex AGREE
2405-24-32	2-11 1/2 NPTF (female) × 1 7/8-12	2.73	2.88	2.2	1.18	1.31	ASSUMED	JIC end: MS33656J==Brennan (VERIFIED); female NPTF: AW tap drill + MS51503 P1 max; L2/hex: Air-Way
2405-32-24	1 1/2-11 1/2 NPTF (female) × 2 1/2-12	2.95	2.63	1.72	1.16	1.78	VERIFIED	RESOLVED 2026-09-06 via Brennan 2405-32-24 drawing (user Chrome): overall 2.95 (AW printed 2.05 - digit misprint), hex 2.625 == AW 2.63, bor
2405-32-32	2-11 1/2 NPTF (female) × 2 1/2-12	2.97	2.88	2.2	1.18	1.78	VERIFIED	MS51503D Amd1 L2/hex AGREE

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2500 90° Union Elbow (JIC 37° male × JIC 37° male)

27 part numbers · TABLED - macro drafted · source table: DesignTable_2500.xlsx

Part number	Thread(s)	M1 (leg 1)	M2 (leg 2)	Y	Bore 1	Bore 2	Status	Source / note
2500-04-04-FG	7/16-20 x 7/16-20 (male JIC both ends)	0.89	0.89	0.44	0.17	0.17	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-04-04	7/16-20 x 7/16-20 (male JIC both ends)	0.89	0.89	0.5	0.17	0.17	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-05-05	1/2-20 x 1/2-20 (male JIC both ends)	0.95	0.95	0.56	0.23	0.23	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-06-04	9/16-18 x 7/16-20 (male JIC both ends)	1.08	0.95	0.63	0.3	0.17	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-06-06	9/16-18 x 9/16-18 (male JIC both ends)	1.08	1.06	0.63	0.3	0.3	ASSUMED	non-FG legs 1.08/1.06 vs J514 M family 1.06 - forged/bar pattern, as-printed
2500-06-06-FG	9/16-18 x 9/16-18 (male JIC both ends)	1.06	1.06	0.56	0.3	0.3	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-08-06	3/4-16 x 9/16-18 (male JIC both ends)	1.25	1.15	0.81	0.39	0.3	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-08-08-FG	3/4-16 x 3/4-16 (male JIC both ends)	1.25	1.25	0.75	0.39	0.39	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-08-08	3/4-16 x 3/4-16 (male JIC both ends)	1.25	1.25	0.81	0.39	0.39	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-10-08-FG	7/8-14 x 3/4-16 (male JIC both ends)	1.45	1.33	0.88	0.48	0.39	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-10-08	7/8-14 x 3/4-16 (male JIC both ends)	1.45	1.31	0.94	0.48	0.39	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-10-10	7/8-14 x 7/8-14 (male JIC both ends)	1.45	1.45	0.94	0.48	0.48	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-12-08	1 1/16-12 x 3/4-16 (male JIC both ends)	1.66	1.4	1.13	0.61	0.39	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-10-10-FG	7/8-14 x 7/8-14 (male JIC both ends)	1.45	1.45	0.88	0.48	0.48	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-12-10	1 1/16-12 x 7/8-14 (male JIC both ends)	1.66	1.55	1.13	0.61	0.48	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-12-12-FG	1 1/16-12 x 1 1/16-12 (male JIC both ends)	1.66	1.66	1.06	0.61	0.61	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-12-12	1 1/16-12 x 1 1/16-12 (male JIC both ends)	1.66	1.66	1.13	0.61	0.61	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-16-12-FG	1 5/16-12 x 1 1/16-12 (male JIC both ends)	1.81	1.76	1.31	0.84	0.61	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-16-12	1 5/16-12 x 1 1/16-12 (male JIC both ends)	1.81	1.78	1.38	0.84	0.61	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending

Part number	Thread(s)	M1 (leg 1)	M2 (leg 2)	Y	Bore 1	Bore 2	Status	Source / note
2500-16-16-FG	1 5/16-12 x 1 5/16-12 (male JIC both ends)	1.81	1.81	1.31	0.84	0.84	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-16-16	1 5/16-12 x 1 5/16-12 (male JIC both ends)	1.81	1.81	1.38	0.84	0.84	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-20-16	1 5/8-12 x 1 5/16-12 (male JIC both ends)	2.06	2	1.75	1.08	0.84	ASSUMED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-20-20-FG	1 5/8-12 x 1 5/8-12 (male JIC both ends)	2.06	2.06	1.63	1.08	1.08	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-20-20	1 5/8-12 x 1 5/8-12 (male JIC both ends)	2.06	2.06	1.75	1.08	1.08	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-24-24	1 7/8-12 x 1 7/8-12 (male JIC both ends)	2.33	2.33	2	1.31	1.31	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-32-32-FG	2 1/2-12 x 2 1/2-12 (male JIC both ends)	3.06	3.06	2.5	1.78	1.78	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending
2500-24-24-FG	1 7/8-12 x 1 7/8-12 (male JIC both ends)	2.33	2.33	1.88	1.31	1.31	VERIFIED	Air-Way s117 (double-pass 27 rows; survey said 28 - page wins); matched legs == SAE J514 M family via MS51510E; MS51505 direct check pending

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2502 90° Female Pipe Elbow (JIC 37° male × NPTF female)

29 part numbers · TABLED · source table: DesignTable_2502.xlsx

Part number	Thread(s)	M1	N1	Y1	Bore	Tap drill Ø	Status	Source / note
2502-04-02-FG	7/16-20 (JIC 37) x 1/8-27 (NPTF female)	1.08	0.66	0.56	0.17	0.33	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-04-02	7/16-20 (JIC 37) x 1/8-27 (NPTF female)	1.08	0.66	0.63	0.17	0.33	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-04-04-FG	7/16-20 (JIC 37) x 1/4-18 (NPTF female)	1.22	0.88	0.75	0.17	0.43	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-04-04	7/16-20 (JIC 37) x 1/4-18 (NPTF female)	1.14	0.88	0.75	0.17	0.43	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-04-08	7/16-20 (JIC 37) x 1/2-14 (NPTF female)	1.33	1.23	1.13	0.17	0.7	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-06-02	9/16-18 (JIC 37) x 1/8-27 (NPTF female)	1.08	0.66	0.63	0.3	0.33	ASSUMED	non-FG row vs MS51506 A6-2: AW 1.08/0.66/0.63 vs MS 1.06/0.64/0.562 - forged/bar-stock geometry differs, as-printed
2502-06-04-FG	9/16-18 (JIC 37) x 1/4-18 (NPTF female)	1.23	0.88	0.75	0.3	0.43	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-06-04	9/16-18 (JIC 37) x 1/4-18 (NPTF female)	1.23	0.88	0.75	0.3	0.43	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-06-06-FG	9/16-18 (JIC 37) x 3/8-18 (NPTF female)	1.31	1.02	0.88	0.3	0.57	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-06-06	9/16-18 (JIC 37) x 3/8-18 (NPTF female)	1.3	1.02	0.88	0.3	0.57	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-06-08	9/16-18 (JIC 37) x 1/2-14 (NPTF female)	1.42	1.23	1.13	0.3	0.7	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-08-04	3/4-16 (JIC 37) x 1/4-18 (NPTF female)	1.25	0.91	0.81	0.39	0.43	ASSUMED	non-FG row vs MS51506 A8-4: AW 1.25/0.91/0.81 vs MS 1.25/0.89/0.75 - forged/bar-stock geometry differs, as-printed
2502-08-06-FG	3/4-16 (JIC 37) x 3/8-18 (NPTF female)	1.42	1.02	0.88	0.39	0.57	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-08-06	3/4-16 (JIC 37) x 3/8-18 (NPTF female)	1.42	1.02	0.88	0.39	0.57	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-08-08-FG	3/4-16 (JIC 37) x 1/2-14 (NPTF female)	1.52	1.23	1.06	0.39	0.7	ASSUMED	legs diverge from MS51506 A8-8: AW M1=1.52/N1=1.23 vs MS 1.38/1.23 - AW stands, Brennan tiebreak pending
2502-08-08	3/4-16 (JIC 37) x 1/2-14 (NPTF female)	1.55	1.23	1.13	0.39	0.7	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-08-12	3/4-16 (JIC 37) x 3/4-14 (NPTF female)	1.67	1.36	1.38	0.39	0.91	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-10-08	7/8-14 (JIC 37) x 1/2-14 (NPTF female)	1.64	1.23	1.13	0.48	0.7	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-10-08-FG	7/8-14 (JIC 37) x 1/2-14 (NPTF female)	1.64	1.23	1.06	0.48	0.7	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)

Part number	Thread(s)	M1	N1	Y1	Bore	Tap drill Ø	Status	Source / note
2502-12-08	1 1/16-12 (JIC 37) x 1/2-14 (NPTF female)	1.76	1.23	1.13	0.61	0.7	ASSUMED	non-FG row vs MS51506 A12-8: AW 1.76/1.23/1.13 vs MS 1.66/1.23/1.062 - forged/bar-stock geometry differs, as-printed
2502-12-12-FG	1 1/16-12 (JIC 37) x 3/4-14 (NPTF female)	1.89	1.36	1.31	0.61	0.91	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-12-12	1 1/16-12 (JIC 37) x 3/4-14 (NPTF female)	1.89	1.36	1.38	0.61	0.91	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-12-16	1 1/16-12 (JIC 37) x 1-11 1/2 (NPTF female)	2.02	1.62	1.63	0.61	1.14	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-16-16-FG	1 5/16-12 (JIC 37) x 1-11 1/2 (NPTF female)	2.17	1.62	1.63	0.84	1.14	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-16-16	1 5/16-12 (JIC 37) x 1-11 1/2 (NPTF female)	2.17	1.62	1.63	0.84	1.14	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-16-20	1 5/16-12 (JIC 37) x 1 1/4-11 1/2 (NPTF female)	2.36	1.7	2	0.84	1.48	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-20-20-FG	1 5/8-12 (JIC 37) x 1 1/4-11 1/2 (NPTF female)	2.33	1.7	1.88	1.08	1.48	VERIFIED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-20-20	1 5/8-12 (JIC 37) x 1 1/4-11 1/2 (NPTF female)	2.33	1.7	2	1.08	1.48	ASSUMED	MS51506F Amd1 corroborates (19-row map); tap drill: Air-Way only (MS has no drill column); JIC end VERIFIED (MS33656J)
2502-24-24	1 7/8-12 (JIC 37) x 1 1/2-11 1/2 (NPTF female)	2.89	2.08	2.5	1.31	1.72	ASSUMED	non-FG row vs MS51506 A24: AW 2.89/2.08/2.5 vs MS 2.89/2.08/2.562 - forged/bar-stock geometry differs, as-printed; Y1 2.50 vs MS51506 A24 Y1

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2503 45° Male Elbow (JIC 37° male × NPTF male)

60 part numbers · TABLED · source table: DesignTable_2503.xlsx

Part number	Thread(s)	M2	N2	Y	Bore	Bore 1	Status	Source / note
2503-02-02-FG	5/16-24 (JIC 37) x 1/8-27 (NPTF)	0.69	0.52	0.44	0.06	0.19	ASSUMED	legs diverge from MS51508 A2: AW M2=0.69/N2=0.52 vs MS 0.66/0.52 - AW stands (dimension authority), Brennan tiebreak pending
2503-02-02	5/16-24 (JIC 37) x 1/8-27 (NPTF)	0.68	0.64	0.5	0.06	0.19	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-03-02-FG	3/8-14 (JIC 37) x 1/8-24 (NPTF)	0.69	0.52	0.44	0.13	0.19	ASSUMED	legs diverge from MS51508 A3: AW M2=0.69/N2=0.52 vs MS 0.66/0.52 - AW stands (dimension authority), Brennan tiebreak pending; page prints B
2503-03-02	3/8-24 (JIC 37) x 1/8-27 (NPTF)	0.75	0.64	0.5	0.13	0.19	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-04-02-FG	7/16-20 (JIC 37) x 1/8-27 (NPTF)	0.72	0.64	0.44	0.17	0.19	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-04-02	7/16-20 (JIC 37) x 1/8-27 (NPTF)	0.72	0.64	0.5	0.17	0.19	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-04-04-FG	7/16-20 (JIC 37) x 1/4-18 (NPTF)	0.82	0.86	0.56	0.17	0.28	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-04-04	7/16-20 (JIC 37) x 1/4-18 (NPTF)	0.81	0.84	0.63	0.17	0.28	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-04-06-FG	7/16-20 (JIC 37) x 3/8-18 (NPTF)	0.85	0.95	0.75	0.17	0.41	ASSUMED	M2 printed "085" - read as 0.85 (missing decimal point)
2503-04-06	7/16-20 (JIC 37) x 3/8-18 (NPTF)	0.76	0.95	0.81	0.17	0.41	ASSUMED	M2 0.76 < FG 0.85 (non-FG usually >= FG) - as-printed
2503-04-08-FG	7/16-20 (JIC 37) x 1/2-14 (NPTF)	1.17	0.89	0.88	0.17	0.52	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-05-02-FG	1/2-20 (JIC 37) x 1/8-27 (NPTF)	0.77	0.64	0.56	0.23	0.19	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-05-02	1/2-20 (JIC 37) x 1/8-27 (NPTF)	0.77	0.64	0.56	0.23	0.19	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-05-04-FG	1/2-20 (JIC 37) x 1/4-18 (NPTF)	0.82	0.88	0.56	0.23	0.28	ASSUMED	legs diverge from MS51508 A5-4: AW M2=0.82/N2=0.88 vs MS 0.82/0.86 - AW stands (dimension authority), Brennan tiebreak pending
2503-05-04	1/2-20 (JIC 37) x 1/4-18 (NPTF)	0.82	0.88	0.63	0.23	0.28	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-05-06	1/2-20 (JIC 37) x 3/8-18 (NPTF)	0.85	0.95	0.81	0.23	0.41	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-02-FG	9/16-18 (JIC 37) x 1/8-27 (NPTF)	0.83	0.67	0.56	0.3	0.19	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-02	9/16-18 (JIC 37) x 1/8-27 (NPTF)	0.83	0.65	0.63	0.3	0.19	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-04-FG	9/16-18 (JIC 37) x 1/4-18 (NPTF)	0.83	0.86	0.56	0.3	0.28	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-04	9/16-18 (JIC 37) x 1/4-18 (NPTF)	0.83	0.86	0.63	0.3	0.3	ASSUMED	D1 printed 0.30 (FG row and MS51508 A6 drill .281 say 0.28) - as-printed, likely typo

Part number	Thread(s)	M2	N2	Y	Bore	Bore 1	Status	Source / note
2503-06-06-FG	9/16-18 (JIC 37) x 3/8-18 (NPTF)	0.91	0.95	0.75	0.3	0.41	ASSUMED	legs diverge from MS51508 A6-6: AW M2=0.91/N2=0.95 vs MS 0.87/0.95 - AW stands (dimension authority), Brennan tiebreak pending
2503-06-06	9/16-18 (JIC 37) x 3/8-18 (NPTF)	0.91	0.95	0.81	0.3	0.41	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-08-FG	9/16-18 (JIC 37) x 1/2-14 (NPTF)	0.88	1.17	0.88	0.3	0.53	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-08	9/16-18 (JIC 37) x 1/2-14 (NPTF)	1.03	1.17	0.94	0.3	0.53	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-06-12	9/16-18 (JIC 37) x 3/4-14 (NPTF)	0.92	1.2	1.13	0.3	0.72	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-04-FG	3/4-16 (JIC 37) x 1/4-18 (NPTF)	0.98	0.95	0.75	0.39	0.28	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-04	3/4-16 (JIC 37) x 1/4-18 (NPTF)	0.98	1.01	0.81	0.39	0.28	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-06-FG	3/4-16 (JIC 37) x 3/8-18 (NPTF)	0.98	0.95	0.75	0.39	0.41	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-06	3/4-16 (JIC 37) x 3/8-18 (NPTF)	0.98	0.95	0.81	0.39	0.41	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-08-FG	3/4-16 (JIC 37) x 1/2-14 (NPTF)	1.11	1.17	0.88	0.39	0.53	ASSUMED	legs diverge from MS51508 A8-8: AW M2=1.11/N2=1.17 vs MS 0.99/1.17 - AW stands (dimension authority), Brennan tiebreak pending
2503-08-08	3/4-16 (JIC 37) x 1/2-14 (NPTF)	1.13	1.17	0.94	0.39	0.53	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-12-FG	3/4-16 (JIC 37) x 3/4-14 (NPTF)	1.04	1.2	1.06	0.39	0.72	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-08-12	3/4-16 (JIC 37) x 3/4-14 (NPTF)	1.19	1.2	1.13	0.39	0.72	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-06-FG	7/8-14 (JIC 37) x 3/8-18 (NPTF)	1.11	0.98	0.88	0.48	0.41	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-06	7/8-14 (JIC 37) x 3/8-18 (NPTF)	1.11	1.06	0.94	0.48	0.41	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-08-FG	7/8-14 (JIC 37) x 1/2-14 (NPTF)	1.11	1.17	0.88	0.48	0.53	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-08	7/8-14 (JIC 37) x 1/2-14 (NPTF)	1.11	1.17	0.94	0.48	0.53	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-12-FG	7/8-14 (JIC 37) x 3/4-14 (NPTF)	1.08	1.16	1.06	0.48	0.72	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-10-12	7/8-14 (JIC 37) x 3/4-14 (NPTF)	1.26	1.2	1.13	0.48	0.64	ASSUMED	D1 printed 0.64 vs FG 0.72 (3/4-14 tap drill is .72) - as-printed, likely typo
2503-12-08-FG	1 1/16-12 (JIC 37) x 1/2-14 (NPTF)	1.28	1.2	1.06	0.61	0.53	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-12-08	1 1/16-12 (JIC 37) x 1/2-14 (NPTF)	1.28	1.19	1.13	0.61	0.53	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-12-12-FG	1 1/16-12 (JIC 37) x 3/4-14 (NPTF)	1.28	1.2	1.06	0.61	0.72	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)

Part number	Thread(s)	M2	N2	Y	Bore	Bore 1	Status	Source / note
2503-12-12	1 1/16-12 (JIC 37) x 3/4-14 (NPTF)	1.28	1.2	1.13	0.61	0.72	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-12-16-FG	1 1/16-12 (JIC 37) x 1-11 1/2 (NPTF)	1.42	1.48	1.31	0.61	0.94	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-14-12-FG	1 3/16-12 (JIC 37) x 3/4-14 (NPTF)	1.45	1.26	1.31	0.72	0.72	ASSUMED	legs diverge from MS51508 A14: AW M2=1.45/N2=1.26 vs MS 1.39/1.27 - AW stands (dimension authority), Brennan tiebreak pending
2503-16-12-FG	1 5/16-12 (JIC 37) x 3/4-14 (NPTF)	1.47	1.29	1.31	0.84	0.72	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-16-12	1 5/16-12 (JIC 37) x 3/4-14 (NPTF)	1.47	1.28	1.38	0.84	0.72	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-16-16	1 5/16-12 (JIC 37) x 1-11 1/2 (NPTF)	1.47	1.48	1.38	0.84	0.94	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-16-20-FG	1 1/16-12 (JIC 37) x 1 1/4-11 1/2 (NPTF)	1.54	1.67	1.63	0.84	1.25	ASSUMED	B printed "1 1/16-12" (a -16 tube takes 1 5/16-12; non-FG row prints it) - as-printed in Threads note
2503-16-20	1 5/16-12 (JIC 37) x 1 1/4-11 1/2 (NPTF)	1.53	1.66	1.75	0.84	1.25	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-16-16-FG	1 5/16-12 (JIC 37) x 1-11 1/2 (NPTF)	1.47	1.48	1.31	0.84	0.94	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-20-16-FG	1 5/8-12 (JIC 37) x 1-11 1/2 (NPTF)	1.59	1.64	1.63	1.08	0.94	ASSUMED	TUBE OD printed "1" (a -20 is 1 1/4; non-FG row prints it) - as-printed
2503-20-16	1 5/8-12 (JIC 37) x 1-11 1/2 (NPTF)	1.59	1.63	1.75	1.08	0.94	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-20-20-FG	1 5/8-12 (JIC 37) x 1 1/4-11 1/2 (NPTF)	1.59	1.67	1.63	1.08	1.25	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-20-20	1 5/8-12 (JIC 37) x 1 1/4-11 1/2 (NPTF)	1.59	1.67	1.75	1.08	1.25	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-20-24	1 5/8-12 (JIC 37) x 1 1/2-11 1/2 (NPTF)	1.63	1.75	2	1.08	1.5	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-24-24-FG	1 7/8-12 (JIC 37) x 1 1/2-11 1/2 (NPTF)	1.78	1.77	1.88	1.31	1.5	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-24-24	1 7/8-12 (JIC 37) x 1 1/2-11 1/2 (NPTF)	1.78	1.77	2	1.31	1.5	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-32-32-FG	2 1/2-12 (JIC 37) x 2-11 1/2 (NPTF)	2.22	2.11	2.5	1.78	1.94	VERIFIED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)
2503-32-32	2 1/2-12 (JIC 37) x 2-11 1/2 (NPTF)	2.22	2.11	2.5	1.78	1.94	ASSUMED	MS51508E Amd1 corroborates FG rows (24-row map); JIC end axials VERIFIED lib-wide (MS33656J)

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

2603 Union Tee (JIC 37° male × 3)

41 part numbers · TABLED · source table: DesignTable_2603.xlsx

Part number	Thread(s)	Leg 1	Leg 2	Leg 3	Y	Bore 1	Bore 2	Bore 3	Status	Source / note
2603-04-04-04-FG	7/16-20 x 7/16-20 x 7/16-20	0.89	0.89	0.89	0.44	0.17	0.17	0.17	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-04-04-04	7/16-20 x 7/16-20 x 7/16-20	0.89	0.89	0.89	0.5	0.17	0.17	0.17	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-04-04-06	7/16-20 x 7/16-20 x 9/16-18	0.95	0.95	1.06	0.63	0.17	0.17	0.3	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-05-05-05-FG	1/2-20 x 1/2-20 x 1/2-20	0.95	0.95	0.95	0.56	0.23	0.23	0.23	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-05-05-05	1/2-20 x 1/2-20 x 1/2-20	0.95	0.95	0.95	0.56	0.23	0.23	0.23	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-06-06-04	9/16-18 x 9/16-18 x 7/16-20	1.08	1.08	0.95	0.63	0.3	0.3	0.17	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-06-06-06-FG	9/16-18 x 9/16-18 x 9/16-18	1.06	1.06	1.06	0.56	0.3	0.3	0.3	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-06-06-06	9/16-18 x 9/16-18 x 9/16-18	1.08	1.08	1.06	0.63	0.3	0.3	0.3	ASSUMED	legs [1.08, 1.08, 1.06] vs MS51510 M 1.06 (all-equal) - AW stands, forged/bar-stock difference pattern
2603-06-06-08	9/16-18 x 9/16-18 x 3/4-16	1.16	1.16	1.25	0.81	0.3	0.3	0.39	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-06-06-10	9/16-18 x 9/16-18 x 7/8-14	1.22	1.22	1.45	0.94	0.3	0.3	0.48	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-06-06	3/4-16 x 9/16-18 x 9/16-18	1.25	1.16	1.16	0.81	0.39	0.3	0.3	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-06-FG	3/4-16 x 3/4-16 x 9/16-18	1.25	1.25	1.14	0.75	0.39	0.39	0.3	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-06	3/4-16 x 3/4-16 x 9/16-18	1.25	1.25	1.16	0.81	0.39	0.39	0.3	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-08-FG	3/4-16 x 3/4-16 x 3/4-16	1.25	1.25	1.25	0.75	0.39	0.39	0.39	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3	Y	Bore 1	Bore 2	Bore 3	Status	Source / note
2603-08-08-08	3/4-16 x 3/4-16 x 3/4-16	1.25	1.25	1.25	0.81	0.39	0.39	0.39	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-10-FG	3/4-16 x 3/4-16 x 7/8-14	1.33	1.33	1.45	0.88	0.39	0.39	0.48	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-10	3/4-16 x 3/4-16 x 7/8-14	1.31	1.31	1.45	0.94	0.39	0.39	0.48	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-08-08-12	3/4-16 x 3/4-16 x 1 1/16-12	1.41	1.41	1.66	1.13	0.39	0.39	0.61	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-10-10-08-FG	7/8-14 x 7/8-14 x 3/4-16	1.45	1.45	1.33	0.88	0.48	0.48	0.39	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-10-10-08	7/8-14 x 7/8-14 x 3/4-16	1.45	1.45	1.31	0.94	0.48	0.48	0.39	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-10-10-10	7/8-14 x 7/8-14 x 7/8-14	1.45	1.45	1.45	0.94	0.48	0.48	0.48	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-10-10-10-FG	7/8-14 x 7/8-14 x 7/8-14	1.45	1.45	1.45	0.88	0.48	0.48	0.48	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-10-12-FG	1 1/16-12 x 7/8-14 x 1 1/16-12	1.66	1.54	1.66	1.06	0.61	0.48	0.61	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-12-08	1 1/16-12 x 1 1/16-12 x 3/4-16	1.66	1.66	1.41	1.13	0.61	0.61	0.39	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-12-10	1 1/16-12 x 1 1/16-12 x 7/8-14	1.66	1.66	1.55	1.13	0.61	0.61	0.48	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-12-12-FG	1 1/16-12 x 1 1/16-12 x 1 1/16-12	1.66	1.66	1.66	1.06	0.61	0.61	0.61	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-12-12	1 1/16-12 x 1 1/16-12 x 1 1/16-12	1.66	1.66	1.66	1.13	0.61	0.61	0.61	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-12-12-16	1 1/16-12 x 1 1/16-12 x 1 5/16-12	1.78	1.78	1.81	1.38	0.61	0.61	0.84	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-16-12-12	1 5/16-12 x 1 1/16-12 x 1 1/16-12	1.81	1.78	1.78	1.38	0.84	0.61	0.61	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-16-16-12	1 5/16-12 x 1 5/16-12 x 1 1/16-12	1.81	1.81	1.78	1.38	0.84	0.84	0.61	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3	Y	Bore 1	Bore 2	Bore 3	Status	Source / note
2603-16-16-12-FG	1 5/16-12 x 1 5/16-12 x 1 1/16-12	1.81	1.81	1.76	1.31	0.84	0.84	0.61	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-14-14-14-FG	1 3/16-12 x 1 3/16-12 x 1 3/16-12	1.8	1.8	1.8	1.31	0.72	0.72	0.72	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-16-16-16-FG	1 5/16-12 x 1 5/16-12 x 1 5/16-12	1.81	1.81	1.81	1.31	0.84	0.84	0.84	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-16-16-16	1 5/16-12 x 1 5/16-12 x 1 5/16-12	1.81	1.81	1.81	1.38	0.84	0.84	0.84	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-16-16-20	1 5/16-12 x 1 5/16-12 x 1 5/8-12	2	2	2.06	1.75	0.84	0.84	1.08	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-20-16-16	1 5/8-12 x 1 5/16-12 x 1 5/16-12	2.06	2	2	1.75	1.08	0.84	0.84	ASSUMED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-20-20-20-FG	1 5/8-12 x 1 5/8-12 x 1 5/8-12	2.06	2.06	2.06	1.63	1.08	1.08	1.08	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-20-20-20	1 5/8-12 x 1 5/8-12 x 1 5/8-12	2.06	2.06	2.06	1.75	1.08	1.08	1.08	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-24-24-24-FG	1 7/8-12 x 1 7/8-12 x 1 7/8-12	2.33	2.33	2.33	1.88	1.31	1.31	1.31	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-24-24-24	1 7/8-12 x 1 7/8-12 x 1 7/8-12	2.33	2.33	2.33	2	1.31	1.31	1.31	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)
2603-32-32-32-FG	2 1/2-12 x 2 1/2-12 x 2 1/2-12	3.06	3.06	3.06	2.5	1.78	1.78	1.78	VERIFIED	MS51510E corroborates matched sizes (13-row, all legs equal +/-.030); JIC ends VERIFIED (MS33656J)

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

5406 Pipe Reducer Bushing (NPTF male × NPTF female)

31 part numbers · TABLED · source table: DesignTable_5406.xlsx

Part number	Thread(s)	L overall	Hex A/F	Tap drill Ø	Status	Source / note
5406-04-02	1/4-18 (male) x 1/8-27 (female) NPTF	0.85	0.63	0.33	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-06-02	3/8-18 (male) x 1/8-27 (female) NPTF	0.85	0.75	0.33	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-06-04	3/8-18 (male) x 1/4-18 (female) NPTF	0.85	0.75	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-08-02	1/2-14 (male) x 1/8-27 (female) NPTF	1.1	0.88	0.33	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-08-04	1/2-14 (male) x 1/4-18 (female) NPTF	1.1	0.88	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-08-06	1/2-14 (male) x 3/8-18 (female) NPTF	1.1	0.88	0.57	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-12-02	3/4-14 (male) x 1/8-27 (female) NPTF	1.17	1.13	0.33	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-12-04	3/4-14 (male) x 1/4-18 (female) NPTF	1.17	1.13	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-12-06	3/4-14 (male) x 3/8-18 (female) NPTF	1.17	1.13	0.57	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-12-08	3/4-14 (male) x 1/2-14 (female) NPTF	1.17	1.13	0.7	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-16-02	1-11 1/2 (male) x 1/8-27 (female) NPTF	1.36	1.38	0.33	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-16-04	1-11 1/2 (male) x 1/4-18 (female) NPTF	1.36	1.38	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-16-06	1-11 1/2 (male) x 3/8-18 (female) NPTF	1.36	1.38	0.57	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-16-08	1-11 1/2 (male) x 1/2-14 (female) NPTF	1.36	1.38	0.7	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-16-12	1-11 1/2 (male) x 3/4-14 (female) NPTF	1.36	1.38	0.91	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-20-04	1 1/4-11 1/2 (male) x 1/4-18 (female) NPTF	1.47	1.75	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-20-06	1 1/4-11 1/2 (male) x 3/8-18 (female) NPTF	1.47	1.75	0.57	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-20-08	1 1/4-11 1/2 (male) x 1/2-14 (female) NPTF	1.47	1.75	0.7	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-20-12	1 1/4-11 1/2 (male) x 3/4-14 (female) NPTF	1.47	1.75	0.91	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated

Part number	Thread(s)	L overall	Hex A/F	Tap drill Ø	Status	Source / note
5406-20-16	1 1/4-11 1/2 (male) x 1-11 1/2 (female) NPTF	1.47	1.75	1.14	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-04	1 1/2-11 1/2 (male) x 1/4-18 (female) NPTF	1.55	2	0.43	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-06	1 1/2-11 1/2 (male) x 3/8-18 (female) NPTF	1.57	2	0.57	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-08	1 1/2-11 1/2 (male) x 1/2-14 (female) NPTF	1.57	2	0.7	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-12	1 1/2-11 1/2 (male) x 3/4-14 (female) NPTF	1.57	2	0.91	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-16	1 1/2-11 1/2 (male) x 1-11 1/2 (female) NPTF	1.57	2	1.14	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-24-20	1 1/2-11 1/2 (male) x 1 1/4-11 1/2 (female) NPTF	1.57	2	1.48	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-32-08	2-11 1/2 (male) x 1/2-14 (female) NPTF	1.75	2.5	0.7	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-32-12	2-11 1/2 (male) x 3/4-14 (female) NPTF	1.75	2.5	0.91	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-32-16	2-11 1/2 (male) x 1-11 1/2 (female) NPTF	1.75	2.5	1.14	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-32-20	2-11 1/2 (male) x 1 1/4-11 1/2 (female) NPTF	1.75	2.5	1.48	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated
5406-32-24	2-11 1/2 (male) x 1 1/2-11 1/2 (female) NPTF	1.75	2.5	1.72	ASSUMED	Air-Way s86 only (double-pass 31 rows); female NPTF per ANSI B1.20.3, tap drill tabulated

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.

6803 Branch Tee (JIC 37° male × JIC 37° male × SAE ORB branch)

34 part numbers · TABLED · source table: DesignTable_6803.xlsx

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread S	Status	Source / note
6803-04-04-04-NWO-FG	7/16-20 x 7/16-20 x 7/16-20	0.89	0.89	1.03	0.56	0.44	0.17	0.17	0.17	0.265	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-04-04-04-NWO	7/16-20 x 7/16-20 x 7/16-20	0.89	0.89	1.03	0.56	0.5	0.17	0.17	0.17	0.265	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-04-04-06-NWO	7/16-20 x 7/16-20 x 9/16-18	0.95	0.95	1.25	0.69	0.63	0.17	0.17	0.3	0.312	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-05-05-05-NWO	1/2-20 x 1/2-20 x 1/2-20	0.95	0.95	1.13	0.63	0.56	0.23	0.23	0.23	0.265	ASSUMED	MS51529 prints boss leg 1.090; commercial 1.13 stands per the 6801 dash-5 tiebreak
6803-06-06-04-NWO	9/16-18 x 9/16-18 x 7/16-20	1.08	1.08	1.09	0.56	0.63	0.3	0.3	0.17	0.265	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread	Status	Source / note
6803-06-06-05-NWO	9/16-18 x 9/16-18 x 1/2-20	1.08	1.08	1.16	0.63	0.63	0.3	0.3	0.23	0.265	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-06-06-06-NWO-FG	9/16-18 x 9/16-18 x 9/16-18	1.06	1.06	1.25	0.69	0.56	0.3	0.3	0.3	0.312	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-06-06-06-NWO	9/16-18 x 9/16-18 x 9/16-18	1.08	1.08	1.25	0.69	0.63	0.3	0.3	0.3	0.312	ASSUMED	legs differ from MS51529: AW [1.08, 1.08]+boss 1.25 vs MS 1.06/1.25
6803-06-06-08-NWO-FG	9/16-18 x 9/16-18 x 3/4-16	1.14	1.14	1.45	0.88	0.75	0.3	0.3	0.39	0.344	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-06-06-08-NWO	9/16-18 x 9/16-18 x 3/4-16	1.15	1.15	1.45	0.88	0.81	0.3	0.3	0.39	0.344	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-08-08-06-NWO	3/4-16 x 3/4-16 x 9/16-18	1.25	1.25	1.34	0.69	0.81	0.39	0.39	0.3	0.312	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread	Status	Source / note
6803-08-08-08-NWO-F3	3/4-16 x 3/4-16 x 3/4-16	1.25	1.25	1.45	0.88	0.75	0.39	0.39	0.39	0.344	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-08-08-08-NWO	3/4-16 x 3/4-16 x 3/4-16	1.25	1.25	1.45	0.88	0.81	0.39	0.39	0.39	0.344	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-08-08-10-NWO	3/4-16 x 3/4-16 x 7/8-14	1.31	1.31	1.7	1	0.94	0.39	0.39	0.48	0.391	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-08-08-12-NWO	3/4-16 x 3/4-16 x 1 1/16-12	1.4	1.4	1.94	0.88	1.13	0.39	0.39	0.61	0.469	ASSUMED	C4 printed 0.88 (branch -12 hex should be 1.25) - as-printed
6803-10-08-10-NWO	7/8-14 x 3/4-16 x 7/8-14	1.45	1.31	1.7	1	0.94	0.48	0.39	0.48	0.391	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-10-10-08-NWO	7/8-14 x 7/8-14 x 3/4-16	1.45	1.45	1.51	0.88	0.94	0.48	0.48	0.39	0.344	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread	Status	Source / note
6803-10-10-10-NWO-FG	7/8-14 x 7/8-14 x 7/8-14	1.45	1.45	1.7	1	0.88	0.48	0.48	0.48	0.391	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-10-10-10-NWO	7/8-14 x 7/8-14 x 7/8-14	1.45	1.45	1.7	1	0.94	0.48	0.48	0.48	0.391	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-10-10-12-NWO-FG	7/8-14 x 7/8-14 x 1 1/16-12	1.54	1.54	1.94	1.25	1.06	0.48	0.48	0.61	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-10-10-12-NWO	7/8-14 x 7/8-14 x 1 1/16-12	1.55	1.55	1.94	1.25	1.13	0.48	0.48	0.61	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-12-06-10-NWO	1 1/16-12 x 9/16-18 x 7/8-14	1.66	1.31	1.8	1	1.13	0.61	0.3	0.48	0.391	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-12-12-10-NWO	1 1/16-12 x 1 1/16-12 x 7/8-14	1.66	1.66	1.8	1	1.13	0.61	0.61	0.48	0.391	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread	Status	Source / note
6803-12-12-12-NWO-FG	1 1/16-12 x 1 1/16-12 x 1 1/16-12	1.65	1.65	1.94	1.25	1.06	0.61	0.61	0.61	0.469	ASSUMED	legs differ from MS51529: AW [1.65, 1.65]+boss 1.94 vs MS 1.66/1.94
6803-12-12-12-NWO	1 1/16-12 x 1 1/16-12 x 1 1/16-12	1.66	1.66	1.94	1.25	1.13	0.61	0.61	0.61	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-12-12-16-NWO	1 1/16-12 x 1 1/16-12 x 1 5/16-12	1.78	1.78	2.05	1.5	1.38	0.61	0.61	0.84	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-16-16-12-NWO	1 5/16-12 x 1 5/16-12 x 1 1/16-12	1.81	1.81	2.06	1.25	1.38	0.84	0.84	0.61	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-16-16-16-NWO-FG	1 5/16-12 x 1 5/16-12 x 1 5/16-12	1.81	1.81	2.05	1.5	1.31	0.84	0.84	0.84	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-16-16-16-NWO	1 5/16-12 x 1 5/16-12 x 1 5/16-12	1.81	1.81	2.05	1.5	1.38	0.84	0.84	0.84	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Part number	Thread(s)	Leg 1	Leg 2	Leg 3 (branch)	Hex A/F	Y	Bore 1	Bore 2	Bore 3	ORB thread	Status	Source / note
6803-20-16-16-NWO	1 5/8-12 x 1 5/16-12 x 1 5/16-12	2.06	2	2.23	1.5	1.75	1.08	0.84	0.84	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-20-20-20-NWO-FG	1 5/8-12 x 1 5/8-12 x 1 5/8-12	2.06	2.06	2.25	1.88	1.63	1.08	1.08	1.08	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-20-20-20-NWO	1 5/8-12 x 1 5/8-12 x 1 5/8-12	2.06	2.06	2.25	1.88	1.75	1.08	1.08	1.08	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-20-20-24-NWO	1 5/8-12 x 1 5/8-12 x 1 7/8-12	2.19	2.19	2.39	2.13	2	1.08	1.08	1.31	0.469	ASSUMED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)
6803-24-24-24-NWO	1 7/8-12 x 1 7/8-12 x 1 7/8-12	2.33	2.33	2.39	2.13	2	1.31	1.31	1.31	0.469	VERIFIED	JIC ends VERIFIED (MS33656J); ORB end sourced; legs: Air-Way (MS51529 corroborates same-dash rows)

Values are the catalog (Air-Way) driving dimensions used in the SolidWorks design table. Standard end geometry (37° seat, relief, thread zone, ORB groove) is applied per dash size from the standards above and is not repeated per row. ASSUMED rows: NOT STANDARDS-VERIFIED — not released for production.